

CT-01

1. True or False?

a) A user requests a Web page that consists of some text and three images. For this page, the client will send one request message and receive four response messages.

False. The client sends one request for the HTML page and separate requests for each image (total 4 requests), receiving 4 responses.

b) Two distinct Web pages (for example, www.mit.edu/research.html and www.mit.edu/students.html) can be sent over the same persistent connection.

True. Persistent connections allow multiple objects to be sent over a single TCP connection.

c) With nonpersistent connections between browser and origin server, it is possible for a single TCP segment to carry two distinct HTTP request messages.

False. Nonpersistent connections handle one request per TCP connection.

d) The Date: header in the HTTP response message indicates when the object in the response was last modified.

False. The Date header indicates when the response was sent. The Last-Modified header shows the last modification date.

e) HTTP response messages never have an empty message body.

False. Some HTTP responses, such as 204 No Content, have empty bodies.

2. What is the HOL blocking issue in HTTP/1.1? How does HTTP/2 attempt to solve it?

HOL (Head-of-line) blocking occurs when a packet at the front of the queue prevents others from being processed. In HTTP/1.1, requests are sent sequentially over a single connection, so one slow response can block others. HTTP/2 solves this using multiplexing: multiple streams (requests/responses) can be sent concurrently over a single TCP connection, reducing delay.

3. What does a stateless protocol mean? Is IMAP stateless? What about SMTP?

A stateless protocol does not retain information about past client requests. Each request is treated as independent.

- IMAP is stateful: it maintains session state across multiple commands.

- SMTP is also stateful during a session (e.g., email sending involves a sequence of commands that build on one another).

4. What is the difference between recursive and iterative DNS queries?

In a recursive DNS query, the DNS server takes full responsibility for resolving the name and returns the final answer to the client.

In an iterative query, the DNS server returns the best answer it can (e.g., a referral to another server), and the client must continue querying.

5.

Consider a client that wants to retrieve an object of size 850,000 bits. The object can either be retrieved directly from the origin server, which is 20,000 km away, or from a cache that is only 1,000 km away. The propagation speed over the link is 2.5×10^8 and the transmission rate is 1 Mbps. Assume that the transfer starts after 1 round-trip time (RTT).

- a) Compute the total time to retrieve the object without a cache.
- b) Compute the total time to retrieve the object with a cache.
- c) Compute the time saved and the percentage of time saved due to caching.

Solution to Web Caching Problem

Given:

- Distance to server = 20,000 km
- Distance to cache = 1,000 km
- Propagation speed = 2.5×10^8 m/s
- Object size = 850,000 bits
- Transmission rate = 1 Mbps
- 1 round-trip time (RTT) before transmission starts

a) Without Cache

- Distance to server = 20,000 km = 20,000,000 meters
- Propagation delay = $20,000,000 / (2.5 \times 10^8) = 0.08$ sec
- RTT = $2 \times 0.08 = 0.16$ sec
- Transmission delay = $850,000 / 1,000,000 = 0.85$ sec
- Total time = RTT + Transmission delay = $0.16 + 0.85 = 1.01$ seconds

b) With Cache (1,000 km away)

- Distance = 1,000 km = 1,000,000 meters
- Propagation delay = $1,000,000 / (2.5 \times 10^8) = 0.004$ sec
- RTT = $2 \times 0.004 = 0.008$ sec
- Transmission delay = 0.85 sec (same size)
- Total time = $0.008 + 0.85 = 0.858$ seconds

c) Time Saved

- Time saved = $1.01 - 0.858 = 0.152$ seconds
- Percentage saved = $(0.152 / 1.01) \times 100 \approx 15.05\%$

CT-02

1. Forwarding is the process of determining the complete path a packet takes from source to destination.

Answer: False

Why: Forwarding moves a packet between interfaces using local routing information, not determining the full path, which is handled by routing.

2. In the data plane, the router decides how to forward a packet based on local information only.

Answer: True

3. A switch fabric connects a router's input ports to its output ports.

Answer: True

4. Longest prefix matching chooses the shortest prefix that matches the destination address.

Answer: False

Why: Longest prefix matching selects the longest matching prefix for the most specific route.

5. Weighted Fair Queuing (WFQ) treats all packet flows equally regardless of priority.

Answer: False

Why: WFQ assigns weights to flows based on priority, prioritizing higher-weighted flows.

6. The "best-effort" service model of the Internet does not guarantee delivery or timing.

Answer: True

7. DHCP can assign IP addresses dynamically and also provide DNS server information.

Answer: True

CT-03

1. CDMA (Code Division Multiple Access) allows multiple users to transmit data simultaneously using different frequency bands.

Answer: False

Why: CDMA uses different codes, not frequency bands, to allow multiple users to transmit data simultaneously.

2. A hidden terminal problem occurs when two stations cannot detect each other but both send data to a common receiver, causing collisions.

Answer: True

3. In 802.11 wireless LAN, collisions only happen during data transmission, not during acknowledgment (ACK).

Answer: False

Why?

While ACKs are typically short and less likely to collide, **collisions can still occur during ACK transmission**. For example:

- If multiple stations receive a frame and try to send an ACK simultaneously, they might interfere.
- Also, if a station sends an ACK while another is transmitting, a collision may occur.

Moreover, **in some cases**, such as when the sender receives a corrupted ACK or no ACK at all, it may retransmit — indicating that ACKs are vulnerable to interference.

4. In symmetric key cryptography, the sender and receiver use different keys for encryption and decryption.

Answer: False

Why: Symmetric key cryptography uses the same key for both encryption and decryption.

5. RSA is considered secure because it is easy to factor large prime numbers.

Answer: False

Why: RSA's security relies on the difficulty of factoring large prime numbers, not its ease.

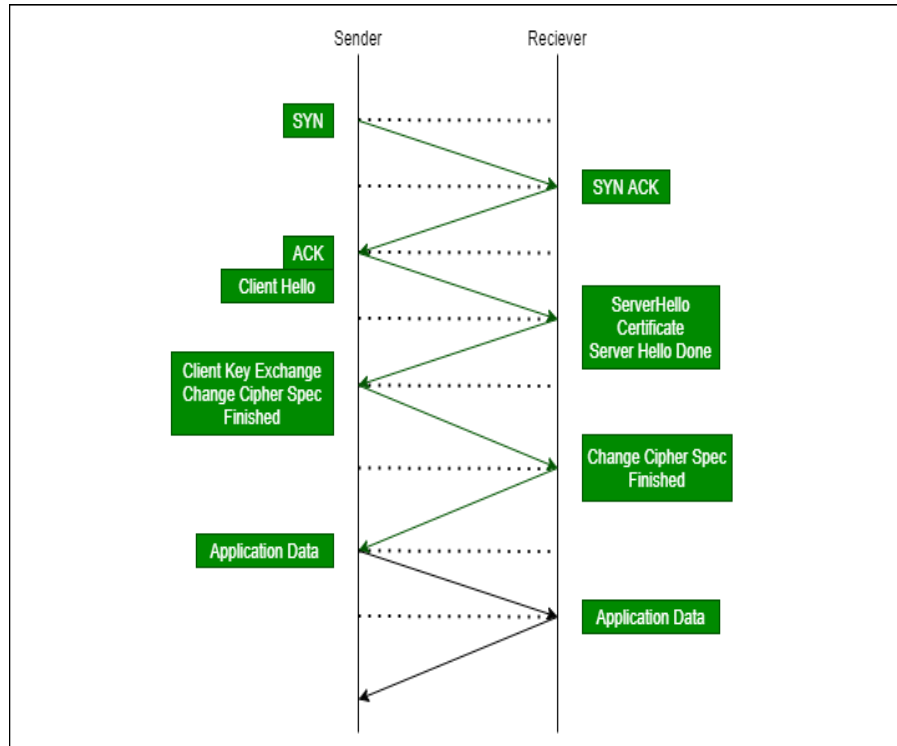
6. TLS provides confidentiality, authentication, and data integrity.

Answer: True

TLS (Transport Layer Security) provides:

- **Confidentiality** via encryption (e.g., AES),
- **Authentication** via digital certificates (server and optionally client),
- **Data integrity** via message authentication codes (MACs) or AEAD ciphers.

1. Draw the initial handshake diagram of TLS Protocol. How many cryptographic keys are needed here?



Cryptographic Keys Involved

1. **Server's private key** – decrypt PreMasterSecret
2. **Server's public key** – encrypt PreMasterSecret
3. **Client-to-server session key** – symmetric key for client messages
4. **Server-to-client session key** – symmetric key for server messages
5. **Optional MAC keys** – for message integrity

Total keys: At least **4 keys** (2 asymmetric, 2 symmetric)

2. What is DES? Explain why 3-DES is secured than DES?

DES (Data Encryption Standard)

DES (Data Encryption Standard) is a symmetric-key block cipher used for encrypting electronic data. It uses the same key for both encryption and decryption.

Encryption Process:

1. Split 64-bit plaintext into two halves.
2. Perform 16 rounds of processing using key schedule.
3. Combine halves to get 64-bit ciphertext.

Security:

- DES was considered secure initially, but 56-bit key is too short for modern brute-force attacks.
- Example: DES Challenge decrypted a message in less than a day using brute force.

3-DES (Triple DES)**Definition:**

- 3-DES enhances DES by applying DES encryption three times with either two or three different keys.

Encryption Process:

1. Encrypt with Key1 (DES)
2. Decrypt with Key2 (DES)
3. Encrypt with Key3 (DES)

Why 3-DES is More Secure:

1. Longer Effective Key Length:
 - Single DES: 56 bits → vulnerable to brute-force
 - 3-DES: up to 168 bits → much harder to break
2. Triple Rounds:
 - Even if one DES is weak, multiple rounds increase complexity.
3. Compatibility:
 - Can still use DES hardware/software with minor modifications.

Drawback:

- Slower than DES due to three rounds of processing, but security is significantly better.

3.What is Hidden Terminal Problem? Man in the Middle attack can be seen in AP-5. Explain how to deal with the attack using digital signatures and sign message digest.

Hidden Terminal Problem

Occurs in wireless networks when **two nodes (say A and C) cannot “hear” each other**, but both want to send data to the same Access Point (AP).This causes **collisions at the AP**, even though A and C are unaware of each other.

- **Solution:**
 - **RTS/CTS mechanism (Request to Send / Clear to Send):**
 1. Node A sends **RTS** to AP before sending data.
 2. AP responds with **CTS**, notifying all nearby nodes.
 3. Other nodes **wait**, preventing collisions.

Man-in-the-Middle (MITM) Attack

- Attacker secretly **intercepts or alters messages** between a client and AP.
- Could steal sensitive info or modify traffic without detection.

Solution Using Digital Signatures:

1. Sender computes **message digest** (hash) of the message.
2. Sender **signs the digest** with their **private key**.
3. Receiver uses sender's **public key** to verify the signature.
4. Ensures:
 - **Authentication:** Confirms sender identity.
 - **Integrity:** Detects if message was modified.
 - **Non-repudiation:** Sender cannot deny sending the message.

Key Advantages:

- MITM cannot forge the signature without the private key.
- Protects messages even if the attacker intercepts traffic.

4. Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the prefix 223.1.17/24. Also suppose that Subnet 1 is required to support at least 60 interfaces, Subnet 2 is to support at least 90 interfaces, and Subnet 3 is to support at least 12 interfaces. Provide three network addresses (of the form a.b.c.d/x) that satisfy these constraints

Ans:

Requirements → VLSM choice

- Subnet 2 needs ≥ 90 hosts → smallest power-of-two block ≥ 90 is 128 → /25 (126 usable).
- Subnet 1 needs ≥ 60 hosts → smallest block ≥ 60 is 64 → /26 (62 usable).
- Subnet 3 needs ≥ 12 hosts → smallest block ≥ 12 is 16 → /28 (14 usable).

Allocate largest first to avoid overlap, all inside 223.1.17.0/24.

Assigned subnets

1. Subnet 2: 223.1.17.0/25
 - Mask: 255.255.255.128
 - Address range: 223.1.17.0 – 223.1.17.127
 - Network ID: 223.1.17.0
 - Broadcast: 223.1.17.127
 - Usable hosts: 223.1.17.1 – 223.1.17.126 (126 hosts)
2. Subnet 1: 223.1.17.128/26
 - Mask: 255.255.255.192
 - Address range: 223.1.17.128 – 223.1.17.191
 - Network ID: 223.1.17.128
 - Broadcast: 223.1.17.191
 - Usable hosts: 223.1.17.129 – 223.1.17.190 (62 hosts)

3. Subnet 3: 223.1.17.192/28

- Mask: 255.255.255.240
- Address range: 223.1.17.192 – 223.1.17.207
- Network ID: 223.1.17.192
- Broadcast: 223.1.17.207
- Usable hosts: 223.1.17.193 – 223.1.17.206 (14 hosts)

Why these fit

- All three blocks are contiguous, non-overlapping, and live within 223.1.17.0/24.
- Host capacities meet or exceed needs: $126 \geq 90$, $62 \geq 60$, $14 \geq 12$.

Free space left (optional use later)

- Remaining addresses: 223.1.17.208 – 223.1.17.255
e.g., next free block begins at 223.1.17.208/28.

Another

Here's a valid VLSM split inside 223.1.17.0/24:

- Subnet 2 (needs ≥ 90 hosts): **223.1.17.0/25** — 128 addresses (126 usable)
- Subnet 1 (needs ≥ 60 hosts): **223.1.17.128/26** — 64 addresses (62 usable)
- Subnet 3 (needs ≥ 12 hosts): **223.1.17.192/28** — 16 addresses (14 usable)

All three are disjoint and lie within 223.1.17.0/24, meeting the host requirements.